



Redox and Equivalent Concept : RACE-01

**Introduction; Classical and Modern Idea of Oxidation and Reduction Reactions**

1. Classify each of the following unbalanced half-reactions as either an oxidation or a reduction:

- (a)  $\text{Cl}_2(\text{g}) \longrightarrow \text{Cl}^-(\text{aq.})$  (b)  $\text{S}_2\text{O}_3^{2-}(\text{aq.}) \longrightarrow \text{S}_4\text{O}_6^{2-}(\text{aq.})$   
 (c)  $\text{NH}_3(\text{g}) \longrightarrow \text{NO}(\text{g})$  (d)  $\text{C}_2\text{O}_4^{2-}(\text{aq.}) \longrightarrow \text{CO}_2(\text{g})$   
 (e)  $\text{Hg}_2^{2+}(\text{aq.}) \longrightarrow \text{Hg}^{2+}(\text{aq.})$  (f)  $\text{Pb}_3\text{O}_4(\text{s}) \longrightarrow \text{PbCl}_2(\text{aq.})$

CRX397

**Redox Reactions in terms of Electron Transfer; Oxidation Number; Oxidising and Reducing Agents**

2. In the reaction,  $2\text{Na}_2\text{S}_2\text{O}_3 + \text{I}_2 = \text{Na}_2\text{S}_4\text{O}_6 + 2\text{NaI}$ ,  $\text{I}_2$  acts as :

- (A) reducing agent (B) oxidising agent  
 (C) oxidising as well as reducing agent (D) none of the above

CRX359

3. In which of the following compound, the oxidation state of sulphur is +7 ?

- (A)  $\text{Na}_2\text{S}_2\text{O}_8$  (B)  $\text{H}_2\text{S}_2\text{O}_7$  (C)  $\text{H}_2\text{SO}_5$  (D) None

CRX398

4. Oxidation number of Fe in  $\text{Fe}_3\text{O}_4$  are :

- (A) +2 and +3 (B) +1 and +2 (C) +1 and +3 (D) +2 and 5

CRX399

5. Which of the following compounds are arranged in increasing oxidation number of S :-

- (A)  $\text{H}_2\text{SO}_3$ ,  $\text{H}_2\text{S}$ ,  $\text{H}_2\text{SO}_4$ ,  $\text{H}_2\text{S}_2\text{O}_3$  (B)  $\text{H}_2\text{S}_2\text{O}_3$ ,  $\text{H}_2\text{SO}_3$ ,  $\text{H}_2\text{S}$ ,  $\text{H}_2\text{SO}_4$   
 (C)  $\text{H}_2\text{S}$ ,  $\text{H}_2\text{SO}_3$ ,  $\text{H}_2\text{SO}_4$ ,  $\text{H}_2\text{S}_2\text{O}_3$  (D)  $\text{H}_2\text{S}$ ,  $\text{H}_2\text{S}_2\text{O}_3$ ,  $\text{H}_2\text{SO}_3$ ,  $\text{H}_2\text{SO}_4$

CRX386

6. The reaction  $\text{H}_2\text{S} + \text{H}_2\text{O}_2 \longrightarrow \text{S} + 2\text{H}_2\text{O}$  manifests

- (A) Oxidising action of  $\text{H}_2\text{O}_2$  (B) Reducing nature of  $\text{H}_2\text{O}_2$   
 (C) Acidic nature of  $\text{H}_2\text{O}_2$  (D) Alkaline nature of  $\text{H}_2\text{O}_2$

CRX387

7. Which of the following can only acts as oxidising agent?

- (A)  $\text{KMnO}_4$  (B)  $\text{K}_2\text{MnO}_4$  (C)  $\text{H}_2\text{O}_2$  (D)  $\text{SO}_2$

CRX388

**Disproportionation; Comproportionation; Intramolecular Redox**

8. Which of the following is not a disproportionation reaction

- (A)  $2\text{CHO}.\text{COOH} + \text{OH}^- \longrightarrow \text{H}_2\text{C}_2\text{O}_4 + \text{HOCH}_2.\text{COOH}$   
 (B)  $\text{Cl}_2 + 2\text{OH}^- \longrightarrow \text{Cl}^- + \text{ClO}^- + \text{H}_2\text{O}$   
 (C)  $2\text{C}_6\text{H}_5\text{CHO} \xrightarrow{\text{Al}(\text{OC}_2\text{H}_5)_3} \text{C}_6\text{H}_5\text{COOCH}_2\text{C}_6\text{H}_5$   
 (D)  $4\text{CrO}_5 + 6\text{H}_2\text{SO}_4 \longrightarrow 2\text{Cr}_2(\text{SO}_4)_3 + 6\text{H}_2\text{O} + 7\text{O}_2$

CRX358

ANSWER KEY

| Question | 1                |                  |                  |                  |                  |                  | 2 | 3 |
|----------|------------------|------------------|------------------|------------------|------------------|------------------|---|---|
| Answer   | (a)<br>Reduction | (b)<br>Oxidation | (c)<br>Oxidation | (d)<br>Oxidation | (e)<br>Oxidation | (f)<br>Reduction | B | D |
| Question | 4                | 5                | 6                | 7                | 8                |                  |   |   |
| Answer   | A                | D                | A                | A                | D                |                  |   |   |



## Redox and Equivalent Concept : RACE-02

**Equivalent Weight and Valency Factor (n-factor)**

1. Equivalent weight of  $\text{NH}_3$  in the change  $\text{N}_2 \rightarrow \text{NH}_3$  is :

(A)  $\frac{17}{6}$  (B) 17 (C)  $\frac{17}{2}$  (D)  $\frac{17}{3}$

CRX355

2. In the reaction,  $2\text{S}_2\text{O}_3^{2-} + \text{I}_2 \rightarrow \text{S}_4\text{O}_6^{2-} + 2\text{I}^-$ , the eq. wt. of  $\text{Na}_2\text{S}_2\text{O}_3$  is equal to its :

(A) Mol. wt. (B) Mol. wt./2 (C) 2 x Mol. wt. (D) Mol. wt./6

CRX356

3. The equivalent weight of  $\text{Na}_2\text{S}_2\text{O}_3$  as reductant in the reaction,  
 $\text{Na}_2\text{S}_2\text{O}_3 + \text{H}_2\text{O} + \text{Cl}_2 \rightarrow \text{Na}_2\text{SO}_4 + 2\text{HCl} + \text{S}$  is :

(A) (Mol. wt.)/1 (B) (Mol. wt.)/2 (C) (Mol. wt.)/6 (D) (Mol. wt.)/8

CRX357

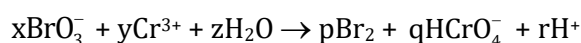
4. Moles of  $\text{K}_2\text{Cr}_2\text{O}_7$  used to oxidise 1 mol  $\text{Fe}_{0.92}\text{O}$  to  $\text{Fe}^{+3}$  are

(A)  $\frac{0.92}{6}$  (B)  $\frac{70}{92} \times \frac{1}{6}$  (C)  $\frac{0.76}{6}$  (D)  $\frac{70}{92} \times \frac{1}{3}$

CRX360

**Balancing of Redox Reactions by Oxidation Number Method**

5. In the ionic reaction,



{x, y, z, p, q, r, are in simplest whole number ratio}

Find the value of (x + y + z) till you get single digit answer :

CRX361

## ANSWER KEY

| Question | 1 | 2 | 3 | 4 | 5 |
|----------|---|---|---|---|---|
| Answer   | D | A | D | C | 2 |



### Redox and Equivalent Concept : RACE-03

#### Gram Equivalents; Normality and Law of Equivalence

- (i) 90 gram  $\text{H}_2\text{C}_2\text{O}_4$  dissolve into 2 litre of solution. Find Normality of solutions.  
 (ii) How much amount of NaOH required to prepare 100 ml of 2N solution.  
 (iii) Determine the number of gram eq. of solute in 250 ml of 0.1N  $\text{Na}_2\text{CO}_3$   
 (iv) Find Normality of 0.2M  $\text{Cr}_2\text{O}_7^{2-}$ , reduced into  $\text{Cr}^{3+}$   
 (v) 49 gram  $\text{H}_2\text{SO}_4$  dissolve in solution to make 10N solution. Find volume of solution. **CRX362**
- If instead of given amount of sodium thiosulphate, 2 moles of sodium thiosulphate along with 3 moles of  $\text{CuSO}_4$  were taken initially. Then moles of sodium cuprothiosulphate formed is -  
 (A) 0 (B) 1 (C) 1.5 (D) 2 **CRX337**
- How many mole of  $\text{FeSO}_4$ ,  $\text{H}_2\text{C}_2\text{O}_4$  and  $\text{FeC}_2\text{O}_4$  are oxidised separately by one mole of  $\text{KMnO}_4$  in acidic medium respectively :-  
 (A) 5 ; 5/2 ; 5/3 (B) 2/5 ; 2 ; 5 (C) 5/3 ; 3/5 ; 2/5 (D) 3/5 ; 2 ; 5 **CRX389**
- How many gram of  $\text{KMnO}_4$  are contained in 4 litre of 0.05 N solution. The  $\text{KMnO}_4$  is to be used as an oxidant in acidic medium ?  
 (A) 1.58 g (B) 15.8 g (C) 6.32 g (D) 31.6 g **CRX390**

#### Redox Titration

- What volume of 0.1 M  $\text{H}_2\text{O}_2$  solution will be required to completely reduce 1 litre of 0.1 M  $\text{KMnO}_4$  in acidic medium.  
 (A) 2500 ml (B) 500 ml (C) 1000 ml (D) 1200 ml **CRX391**
- 10 mole of ferric oxalate is oxidised by x mole of  $\text{MnO}_4^-$  in acidic medium. The value of 'x' is-  
 (A) 12 (B) 4 (C) 40 (D) 18 **CRX392**
- In alkaline medium  $\text{ClO}_2$  oxidises  $\text{H}_2\text{O}_2$  to  $\text{O}_2$  and itself gets reduced to  $\text{Cl}^-$ . How many moles of  $\text{H}_2\text{O}_2$  are oxidised by 1 mole of  $\text{ClO}_2$  ?  
 (A) 1 (B) 1.5 (C) 2.5 (D) 3.5 **CRX393**

#### Acid Base Titration

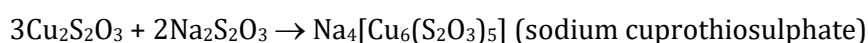
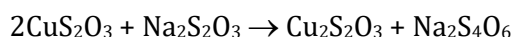
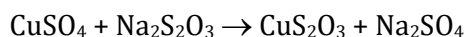
- Problem based on Acid - Base titration  
 (i) 25 ml of N/10 NaOH solution exactly neutralises 20 ml of a monobasic acid solution find Normality of mono basic acid.  
 (ii) How many ml of 1 M  $\text{H}_2\text{SO}_4$  acid solution is required to neutralise 10 ml of 1 M NaOH  
 (iii) 10 ml, 10 M  $\text{H}_2\text{SO}_4$  solution, mixed with 100 ml of NaOH solution. If resultant solution is neutral then find Normality of NaOH solution.  
 (iv) How many moles of NaOH required to neutralise 3 moles of  $\text{KHC}_2\text{O}_4$   
 (v) 45 gram  $\text{H}_2\text{C}_2\text{O}_4$  was dissolved in water and made upto 500 ml. How much volume of this solution required to neutralise 120 ml of N/6 NaOH solution. **CRX363**

9. A sample of  $\text{H}_2\text{SO}_4$  solution having density 1.96g/ml. If 50 ml of this solution is completely neutralized by 2M, 100ml NaOH solution. Wt % of  $\text{H}_2\text{SO}_4$  in the solution is.  
 (A) 6 % (B) 12.5 % (C) 10 % (D) 2 %

CRX394

**Iodimetric and Iodometric Titration****Paragraph for Q.10**

632 g of sodium thiosulphate ( $\text{Na}_2\text{S}_2\text{O}_3$ ) reacts with copper sulphate to form cupric thiosulphate which is reduced by sodium thiosulphate to give cuprous compound which is dissolved in excess of sodium thiosulphate to form a complex compound sodium cuprothiosulphate ( $\text{Na}_4[\text{Cu}_6(\text{S}_2\text{O}_3)_5]$ ).



In this process, 0.2 mole of sodium cuprothiosulphate is formed. (O = 16, Na = 23, S = 32)

10. Moles of sodium thiosulphate reacted and unreacted after the reaction are respectively  
 (A) 3 & 2 (B) 2 & 3 (C) 2.2 & 1.8 (D) 1.8 & 2.2

CRX331

**Back Titration**

11. 4.3 gm of an alkane is burnt in sufficient oxygen. The  $\text{CO}_2$  formed reacts completely with 300 ml, 2N NaOH solution producing  $\text{Na}_2\text{CO}_3$ . The alkane should be  
 (A)  $\text{C}_5\text{H}_{12}$  (B)  $\text{C}_6\text{H}_{12}$  (C)  $\text{C}_6\text{H}_{14}$  (D)  $\text{C}_5\text{H}_{10}$

CRX395

12. An aq. solution of 0.5M  $\text{KMnO}_4$  is divided into two parts. One part of it requires 125 ml of 1.5M aq. solution of oxalate ions in acidic medium, while another part requires 270 ml of 0.5M aq. solution of iodide ions in neutral medium which are converted into  $\text{I}_2$  only. Calculate total volume (mL) of the initial  $\text{KMnO}_4$  solution.

CRX396

**ANSWER KEY**

| Question | 1      |             |             |            |            | 2      | 3 | 4  | 5  | 6   |
|----------|--------|-------------|-------------|------------|------------|--------|---|----|----|-----|
| Answer   | (i) 1N | (ii) 8 gram | (iii) 0.025 | (iv) 1.2 N | (v) 100 ml | A      | A | C  | A  | A   |
| Question | 7      | 8           |             |            |            |        | 9 | 10 | 11 | 12  |
| Answer   | C      | (i) 0.125   | (ii) 5      | (iii) 2    | (iv) 3     | (v) 10 | C | C  | C  | 240 |